ZNU - Collision Detection

1. Theory

Definition 1: A pointers matrix M_{rp} is a collection of vectors drawn from point *p* to each point in rectangular *r*.

Definition 2: A point *p* collides with rectangular *r* if it lays within the rectangular or on any of its sides.

Theory 1: Let M_{rp} be a pointers matrix of point *p* and rectangular *r*, *p* never collides *r* if any of M_{rp} columns is sign-consistent.

Definition 3: A possible collision point (PCP) of rectangular *r*, is a point that is about to, or already did, collide with *r*.

Corollary: The pointers matrix of a PCP, of some rectangular, doesn't contain a sign-consistent column.

Theory 2: Let *p* be a PCP of a rectangular *r*, *p* collides *r* if and only if the circumference of the resulting retangular, after substituting *p* to the closest point to it in *r*, is smaller than or equal to the original one.

2. Implementation

Let p_1, p_2, p_2, ... p_n be arbitrary points in space *s*. And *r* is a rectangular, in *s*, which is drawn from points: r_1, r_2, r_3, and r_4. In order to classify, so to speak, each point whether it's a PCP or not, efficiently. We ought to group all their M_{rp} matrices in one big matrix, that we may call M_r.

$$M_r = \begin{pmatrix} M_{p_1x_1} & M_{p_1y_1} \\ M_{p_1x_2} & M_{p_1y_2} \\ M_{p_1x_3} & M_{p_1y_3} \\ M_{p_1x_4} & M_{p_1y_4} \\ M_{p_2x_1} & M_{p_2y_1} \\ . & \\ . & \\ . & \\ M_{p_nx_4} & M_{p_ny_4} \end{pmatrix}$$

Whereas $M_{rp_ix_k}$ represents the value placed in the first column of the k'th row in the pointers matrix of the i'th point. Note that the *r* has been dropped in the matrix above for simplification.

Next we shall normalize this M_r matrix by dividing it over $|M_r|$, which will leave us with a matrix of positive and negative 1′s. Then our objective is to compact and transform this *4n x 2* matrix into a single vector with n elements, each of which carries the value 0 or 1. If the i'th element value equals 0, then the i'th point may collide *r* (it's considered a PCP), otherwise it should not.

$$(M_r)_{\mathtt{norm}} = \frac{M_r}{\|M_r\|}$$

$$(M_r)_{aggr} = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & \ldots \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & \ldots \\ \ldots & & & & & & & & \\ 0 & 0 & 0 & 0 & \ldots & 1 & 1 & 1 & 1 \end{pmatrix} . (M_r)_{\mathtt{norm}}$$

$$V_r = \left\lfloor \frac{(M_r)_{aggr}}{4} \right\rfloor . \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \\ \ldots \end{pmatrix}$$

Where V_r, with length *n*, classifies points; whether they are PCPs or not. If the i'th value is 0 then p_i is considered a PCP. And note that the floor operation is applied on the matrix elementwise.

Let $c_1, c_2, c_3, \ldots c_k$ be the extracted PCPs of rectangular r from the previous step, and V_x and V_y be the first and second columns in M_{rc_i}, where $i \in \{1, 2, \ldots, k\}$.

$$V_l = (V_x \odot V_x) + (V_y \odot V_y)$$
$$s(V_l) = \text{index of the smallest row value}$$

Recall that M_{rc_i} is being calculated from subtracting the point c_i from the matrix representation of r:

$$r = \begin{pmatrix} r_{x1} & r_{y1} \\ r_{x2} & r_{y2} \\ r_{x3} & r_{y3} \\ r_{x4} & r_{y4} \end{pmatrix}$$

Where each row in the matrix has the x and y values of one point of the rectangular. And the points are sorted, in the matrix, in a way that if each point get subtracted from the one that follows it, and the last point get subtracted from the first one, we shall wind up with a collection of four vectors that form a valid rectangular together.

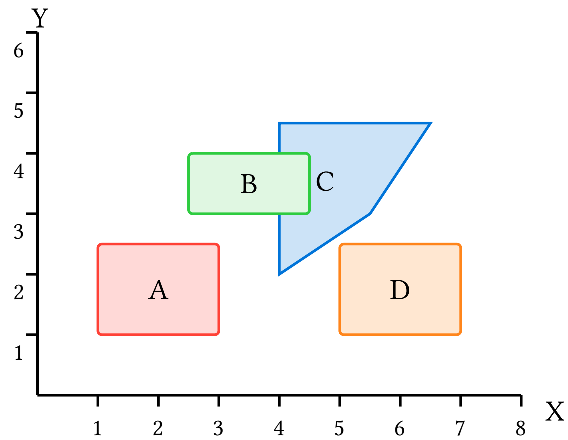
Now, if we substitute the point at index $s(V_l)$ by point c_i, we should wind up with a different rectangular that has a larger circumference if and only if c_i does not collide r.

$$r_{c_i} = \begin{pmatrix} r_{x1} & r_{y1} \\ c_{i_x} & c_{i_y} \\ r_{x3} & r_{y3} \\ r_{x4} & r_{y4} \end{pmatrix}$$

If $circ(r) - circ\left(r_{c_i}\right) \geq 0$, then the point c_i collides rectangular r.

3. Example

Now, let's see it in action. We will draw a simple example here that consists of four rectangulars: A, B, C, and D. Then we will detect collisions to C using the aforementioned approach.



$$r_a = \begin{pmatrix} 1 & 1 \\ 1 & 2.5 \\ 3 & 2.5 \\ 3 & 1 \end{pmatrix} \qquad r_b = \begin{pmatrix} 2.5 & 3 \\ 2.5 & 4 \\ 4.5 & 4 \\ 4.5 & 3 \end{pmatrix}$$

$$r_c = \begin{pmatrix} 4 & 2 \\ 4 & 4.5 \\ 6.5 & 4.5 \\ 5.5 & 3 \end{pmatrix} \qquad r_d = \begin{pmatrix} 5 & 1 \\ 5 & 2.5 \\ 7 & 2.5 \\ 7 & 1 \end{pmatrix}$$

Let P (Points) be an $n \times 2$ matrix that contains all the points in the space, except the rectangular c ones. And then let P_d (Points Duplicated) be a $4n \times 2$ matrix of all the points duplicated three times as elaborated in the sections above.

$$M_{r_c} = \begin{pmatrix} 4 & 2 \\ 4 & 4.5 \\ 6.5 & 4.5 \\ 5.5 & 3 \\ \vdots & \\ 5.5 & 3 \end{pmatrix} - P_d = \begin{pmatrix} 3 & 1 \\ 3 & 3.5 \\ 5.5 & 3.5 \\ 4.5 & 2 \\ \vdots & \end{pmatrix}$$

Note: it's starting from point (1, 1) in r_a.

For the sake of readability, in this document, let's examine only the parts of the matrix appertain to the three, obvious, PCP points on the diagram above.

$$r_{b3} = \begin{pmatrix} -0.5 & -2 \\ -0.5 & 0.5 \\ 2 & 0.5 \\ 1 & -1 \end{pmatrix} \qquad r_{b4} = \begin{pmatrix} -0.5 & -1 \\ -0.5 & 1.5 \\ 2 & 1.5 \\ 1 & 0 \end{pmatrix}$$

$$r_{d2} = \begin{pmatrix} -1 & -0.5 \\ -1 & 2 \\ 1.5 & 2 \\ 0.5 & 0.5 \end{pmatrix}$$

It's clear that each point of these three is considered PCP according to *theory 1*; no single column, of the above matrices, is sign-consistent.

After normalizing, aggregating, dividing by 4, and then apply dot product from the right by $\begin{pmatrix} 1 & 1 \end{pmatrix}$. We get a single vector concluding that only three points are considered PCPs: r_{b3}, r_{b4}, and r_{d2}.

Lastly, we shall apply the circumference method to only these three points in order to determine if eachone does collide or not with r_c.

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